# QC Story Project

# W112 – AT vehicle capacity enhancement (100%)

# QC STORY PROJECT

## PROJECT TITLE :-

To improve the W112 – AT vehicle capacity enhancement (100%) at chassis line

**Mahindra and Mahindra Ltd, Nasik - Plant, Multi- Model Assembly Shop**

| | Plan | Actual |
|---|---|---|
| Start Date | **23.07.19** | **24.07.19** |
| Completion Date | **27.12.19** | **27.12.19** |

Current Status : **Closed**

**Token No : 23061163**

**Name : Nikhil Hatedkar**

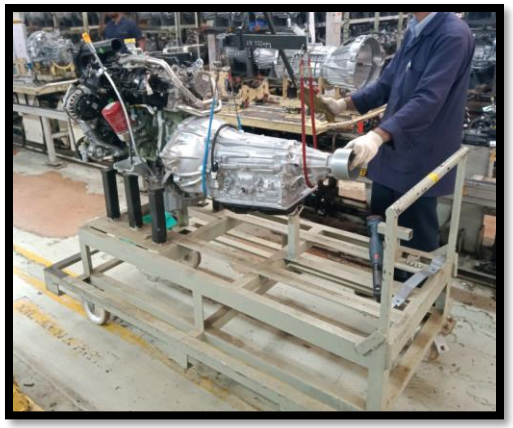
# Cross Functional Team

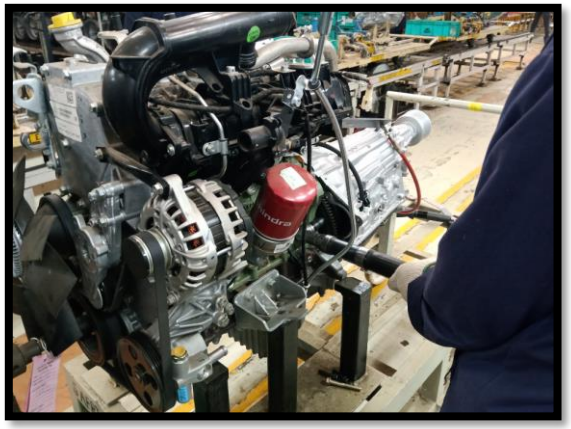
| S. No | Description | Name | Designation | Dept | Phone | Email |
|---|---|---|---|---|---|---|
| 1. | Team Leader | Nikhil Hatedkar | Manager | Mfg | 9421191864 | HATEDKAR.NIKHIL@Mahindra.com |
| 2. | Member 1 | Rohit Ghuge | Manager | Plant Engineering | 9850848696 | GHUGE.ROHIT@mahindra.com |
| 3. | Member 2 | Tushar Patil | Manager | Maintenance | 9922409136 | PATIL.TUSHAR@mahindra.com |
| 4. | Member 3 | Rajesh Dhanokar | Manager | IED | 9730074163 | Dhanokar.Rajesh@Mahindra.com |
| 5. | Member 4 | Jitendra Bhambre | ASSISTANT MANAGER | Maintenance | 8975856767 | BHAMARE.JITENDRA@mahindra.com |
| 6. | Member 5 | Pravin Patil | Manager | TOOLING-TCF | 8888897850 | PATIL.PRAVIN2@mahindra.com |

# Step1. Select Subject

**Brief description of Problem :**

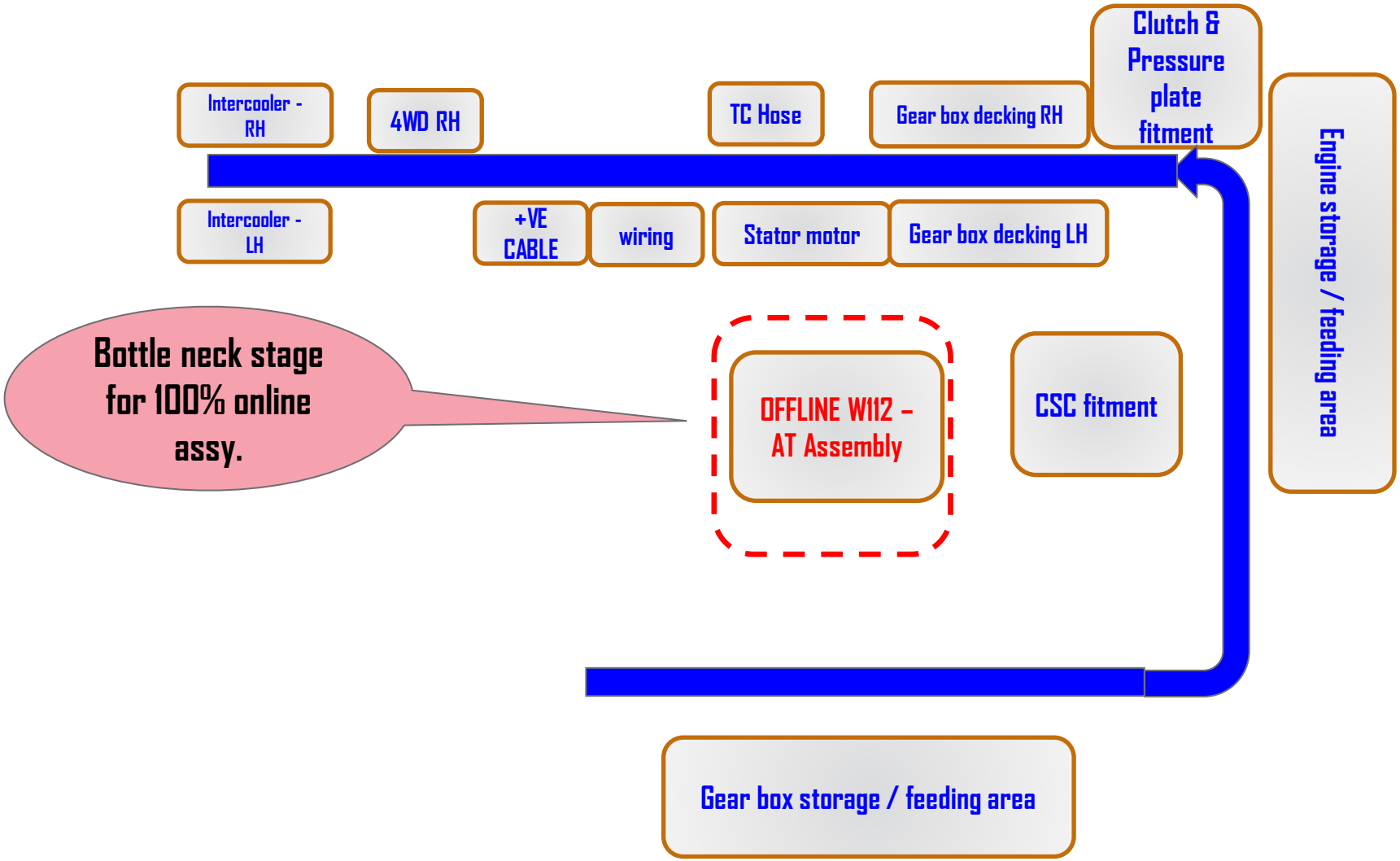
## ENGINE PREPARATION STAGE DEBOTTLENECK FOR W112 AT ONLINE ASSEMBLY





# Step 3. Example Understanding Current Situation

## ENGINE PREPARATION STAGE DEBOTTLENECK FOR W112 AT ONLINE ASSEMBLY



# Step1. Select Subject

**What** : W112 AT Sub assembly stage bottle neck for 100% online assembly.

**Where** : STCF, Engine Preparation stage

**When** : From VP - 0 Phase of project development.

**Why** : Current operation is offline & requires to put additional man power (export manpower) for Sub assly preparation

**Who** : By stage associate while assembly.

**How** : 100 % W112 AT model from inception.

# 2. Justify the choice (reasons)

**PROMISE 2022**
AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment
'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV
Doubling volume from overseas markets

Leading in next generation mobility solutions

**#InspiringPeopleToRise**

Doubling Volumes from Overseas market

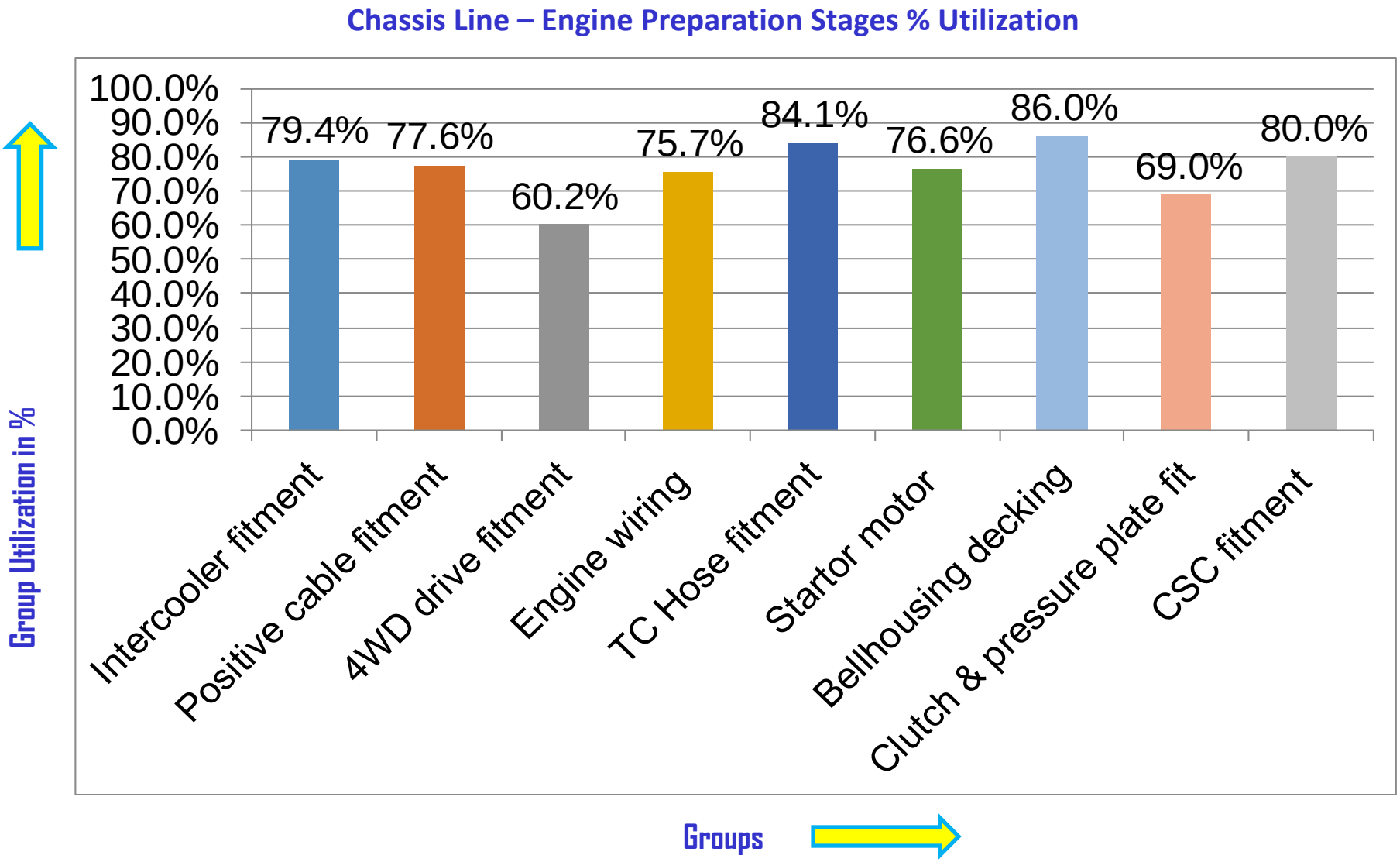
**Departmental Objective :-**
**W112 AT vehicle productivity improvement**

# Step 2. b Immediate Containment Action

- Torque convertor plate pre-fitment at offline
- Torque convertor plate tightening with indexing machine at offline stage
- Gear box loading on engine trolley
- Gear box decking with engine at offline
- Bellhousing bolt tightening at offline stage
- Engine loading on conveyor

**after that preassemble engine given to online for further assy.**

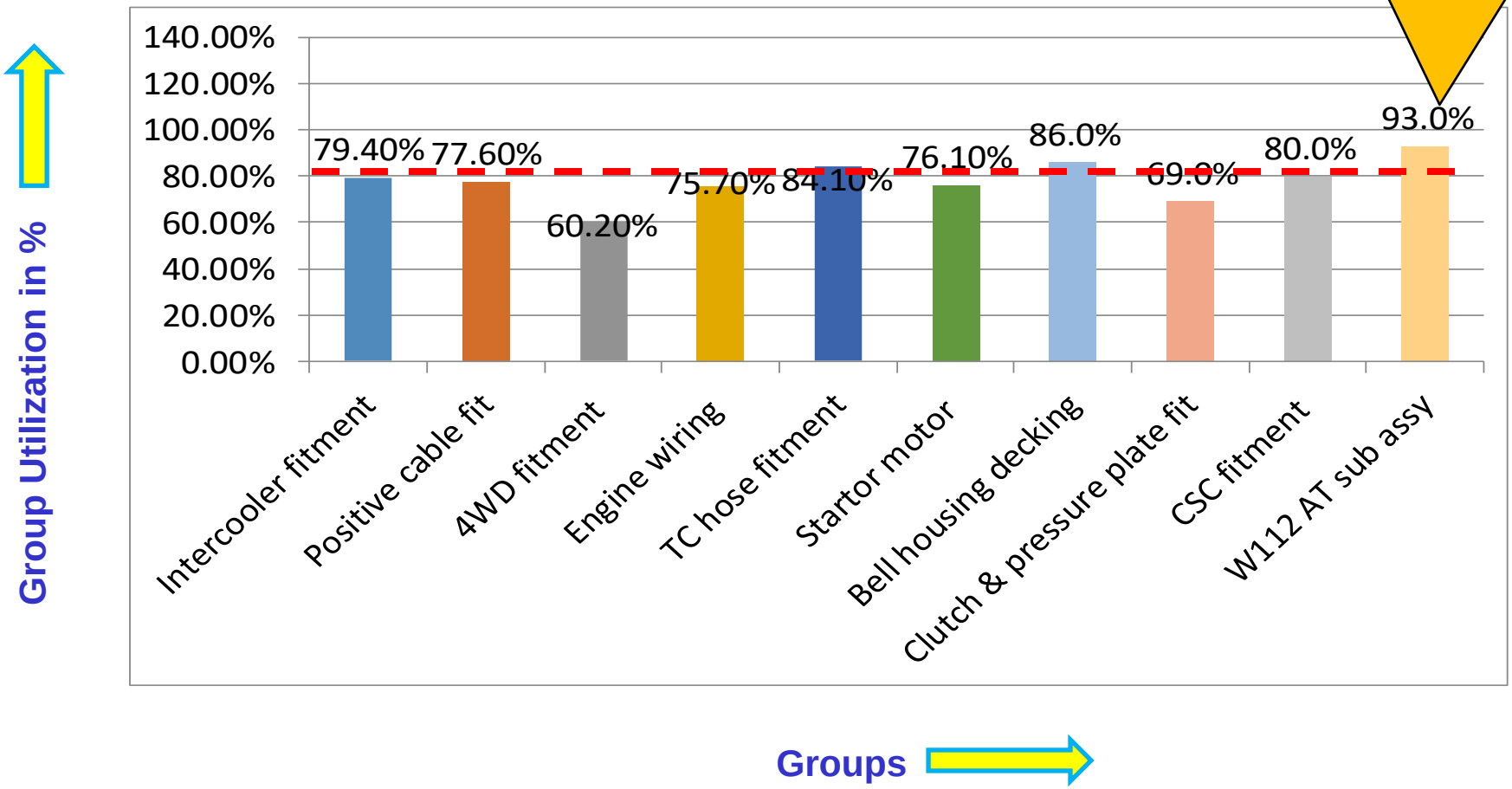
# Step 3. Understanding Current Situation



# Step 3. Understanding Current Situation

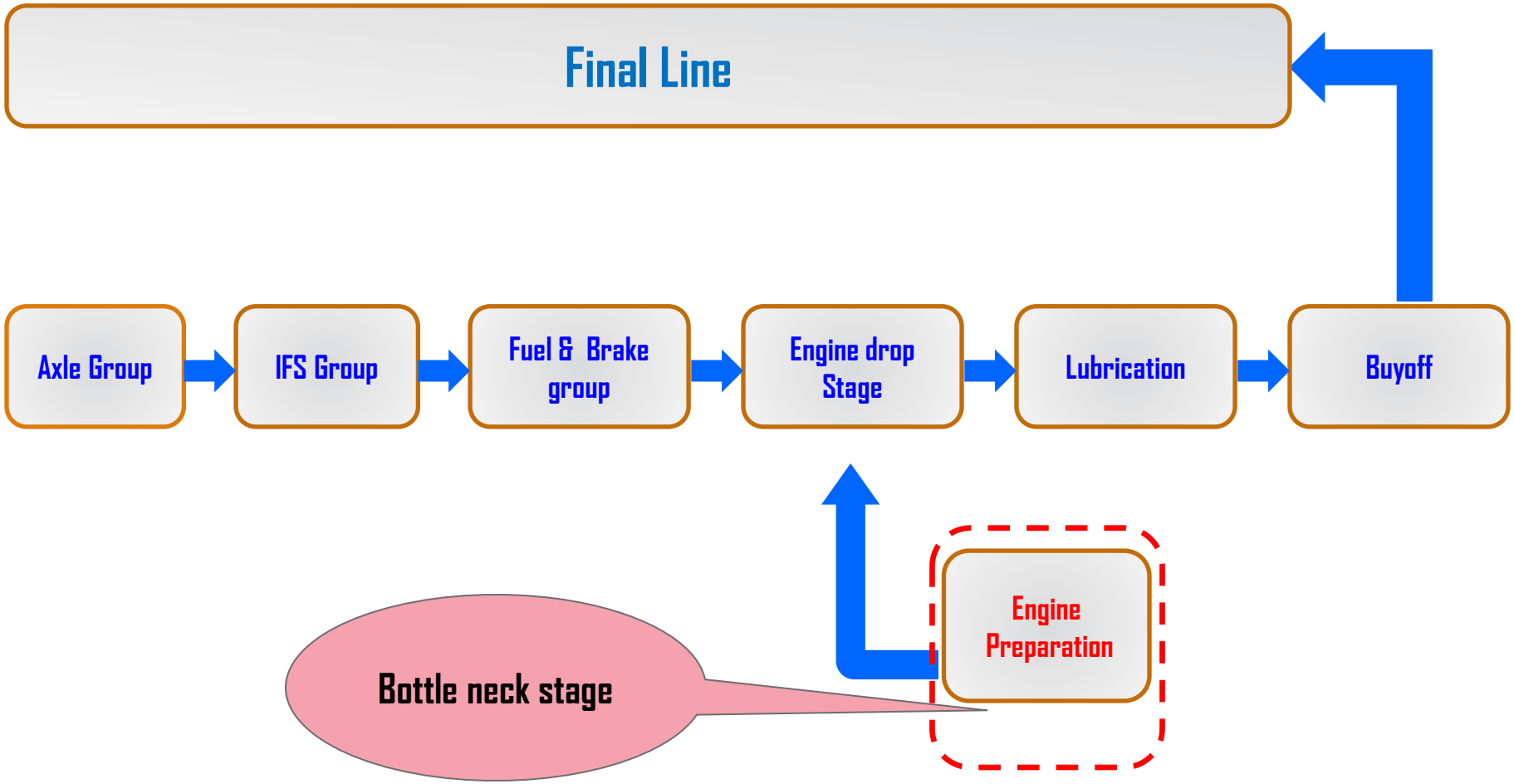
**Addition of**
1. Flex plate fitment using indexing fixture
2. Gear box decking with engine
3. Bellhousing bolt tightening
4. Engine loading on engine conveyor

**Chassis line Utilization __ with W112 AT vehicle**



# Step 3. Example Understanding Current Situation

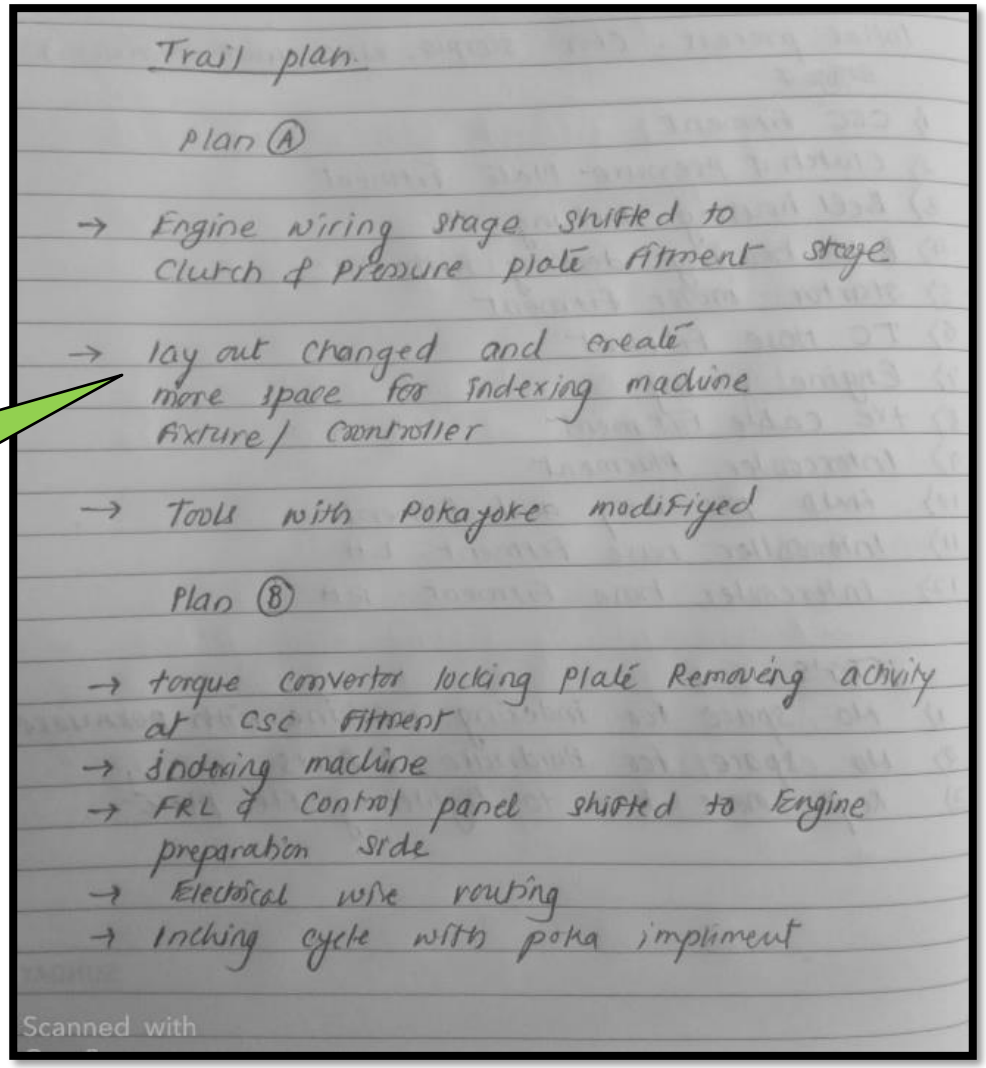
## Chassis line - Process Flow Diagram with W112 AT vehicle



# Step 3. Understanding Current Situation

**Initial Trial findings**

**No space for fitment Front suspension**



Trail plan

Plan (A)

→ Engine wiring stage shifted to Clutch & Pressure plate fitment stage

→ lay out changed and create more space for Indexing machine fixture / Controller

→ Tools with Pokayoke modifiyed

Plan (B)

→ torque convertor locking plate Removing activity at CSC fitment
→ indexing machine
→ FRL & Control panel shifted to Engine preparation side
→ Electrical wire routing
→ Inching cycle with poka impliment

Scanned with

# Step 3. Understanding Current Situation

**Options Workout**

**Workouts by CFT for Addition of Parts at IFS**

Intial process. (For Scorpio, W110, 4WD, Marazzo)

~~Stage 1~~

1) CSC Fitment
2) Clutch & Pressure Plate Fitment
3) Bell housing decking LH
4) Bell housing decking ~~with~~ RH
5) Starter motor Fitment
6) TC Hose Fitment
7) E[illegible]ne wiring
[illegible]e cable Fitment
[illegible]) Intercooler Placment
10) 4WD pre-assy and Fitment
11) Intercooller hose Fitment - LH
12) Intercooler hose Fitment - RH

FTG's

1) No space for indexing machine with POKAYOKE
2) No space for Hardware & Parts
3) Req$^{n}$ more time to tightening Flex plate

# Step 3. Understanding Current Situation

**Options Workout**

**Model wise**
**1.Part fitments**
**2.Process flow**

| Stages | W110/V321 | | W112 |
|---|---|---|---|
| CSC Fitment | – | ✓ | ✗ |
| Bell housing decking | – | ✓ | ✗ |
| Starter motor | – | ✓ | ✓ |
| Engine wiring | – | ✓ | ✓ |
| TC Hose | – | ✓ | ✓ |
| +ve cable | – | ✓ | ✓ |
| Intercooler LH | – | ✓ | ✓ |
| Intercooler RH | — | ✓ | ✓ |

# Step 3. Understanding Current Situation

## Work content study - Engine Preparation stage

| | Engine Preparation | |
|---|---|---|
| 2 | Traveller card preparation for Engine group | 0.23 |
| 3 | Traveller card filling for Engine group | 0.41 |
| | TRANSMISSION MTG 4 BOLT TIGHTENING SCANNING | 0.07 |
| | ENGINE MOUNT 1 BOLT TIGHTENING SCANNING | 0.07 |
| | CSC POKAYOKE SCANNING | 0.07 |
| | ENGINE PLACING ON ENGINE CONVEYOR FIXTURE LH SIDE | 0.04 |
| | STUDS FITMENT ON ENGINE FOR GEAR BOX DECKING | 0.51 |
| | SLEEVE INSERTION AND REMOVAL FOR CENTER BEARING PRESSING | 0.06 |
| 10 | Engine docking with transmission. | 2.54 |
| 11 | STARTER MOTOR FITMENT ON BELL HOUSING | 0.68 |
| | BARCODE SCANNING FOR TRACEBILITY (Starter Motor ) | 0.19 |
| | BARCODE SCANNING FOR TRACEBILITY ( TRANSMISSION ) | 0.19 |
| 12 | Engine wiring fitment on engine | 1.26 |
| | BARCODE SCANNING FOR TRACEBILITY (Engine Wiring ) | 0.19 |
| W105 | Additional clip fitment on engine and transmission during Engine wiring fitment | 0.07 |
| New | + VE WIRING ITMENT ON STATOR & ALTERNATOR | 1.02 |
| 15 | TC HOSE FITMENT ON TURBOCHARGER | 0.47 |
| 16 | Heater hose inlet fitment on engine | 0.31 |
| 17 | Heater hose outlet fitment on engine | 0.11 |
| 18 | Power strg. suction Hose Fitment on Power Strg Pump | 0.58 |
| 20 | Oil separator hose fitment & Insertion of hose vaccum-pressure modulator on intake. | 0.20 |
| 24 | Material Collection | 0.08 |
| 25 | Gear box Insulater bolt Fitment. | 0.41 |
| 29 | Inter Cooler inlet hose | 0.27 |
| 30 | Engine water outlet pipe ( Radiator Hose ) | 0.16 |
| 31 | Inter Cooler outlet hose | 0.28 |
| 32 | Inter Cooler Fitment | 1.01 |
| 41 | Steering reservior fitment | 0.19 |
| 64 | VACUUM HOSE FITMENT | 0.18 |
| U321 | CSC TIGHTENING MULTISPINDLE ROTATION FOR U321 | 0.09 |
| W112_AT | GEAR BOX LOCKING PATTI REMOVAL _AT | 0.28 |
| W112_AT | FLEX PLATE TO TORQUE CONVERTER 6 BOLT FITMENT AND TIGHTENIN | 0.83 |
| W112_AT | WATER COOLING LINE CP CLAMP CONNECTION 2 NOS WITH ENGINE _A | 0.30 |
| W112_AT | BREATHER PIPE ROUTINE AND CLIPPING ON METALLIC PIPE_AT | 0.02 |
| W112_AT | OIL COOLING LINE FITMENT ON TRANSSION 2 BENJO BOLT LOOSE FITMENT_AT | 0.74 |
| W112_AT | AT TRANSMISSION WIRING CLIPPING BKT_AT | 0.31 |
| W112_AT | AT TRANSMISSION WIRING FITMENT LH SIDE_AT | 0.16 |
| W112_AT | STARTER MOTOR DUST SHIELD ADDITIONAL PART FITMENT_AT | 0.02 |
| | **Engine Preparation** | **14.62** |

This assy was done Offline _ Need 2 men with Low utilization of 52% (93/2)

# Step 3. Understanding Current Situation

## Observations from above study

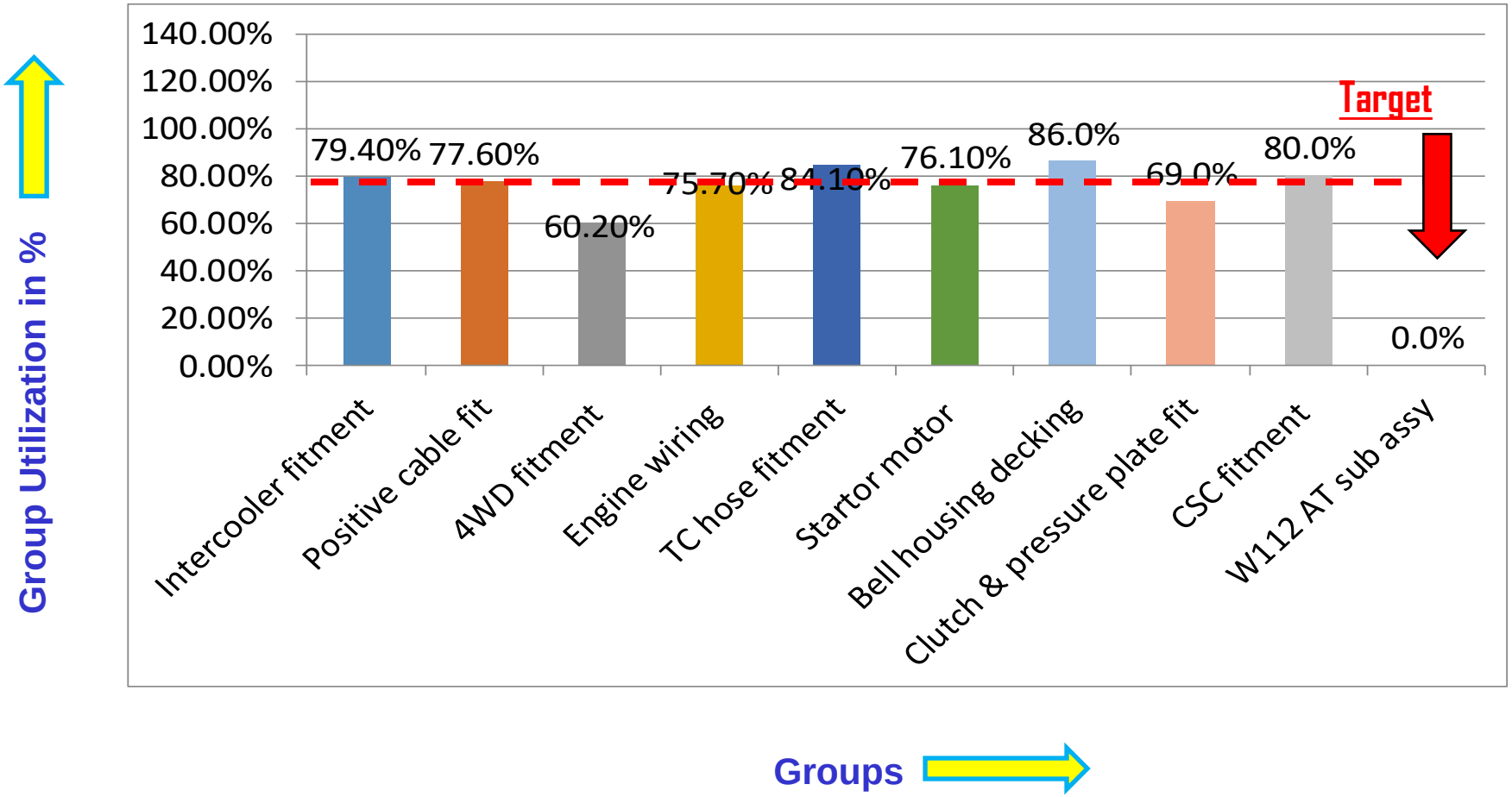
1. Before – This assy. was done Offline Need 2 men with Low utilization of 52%

2. Scope for internal group Line balancing…

3. Activity Deletion or Shifting…

# Step 4. Select Target

**Chassis line Utilization __ with W112 AT vehicle**



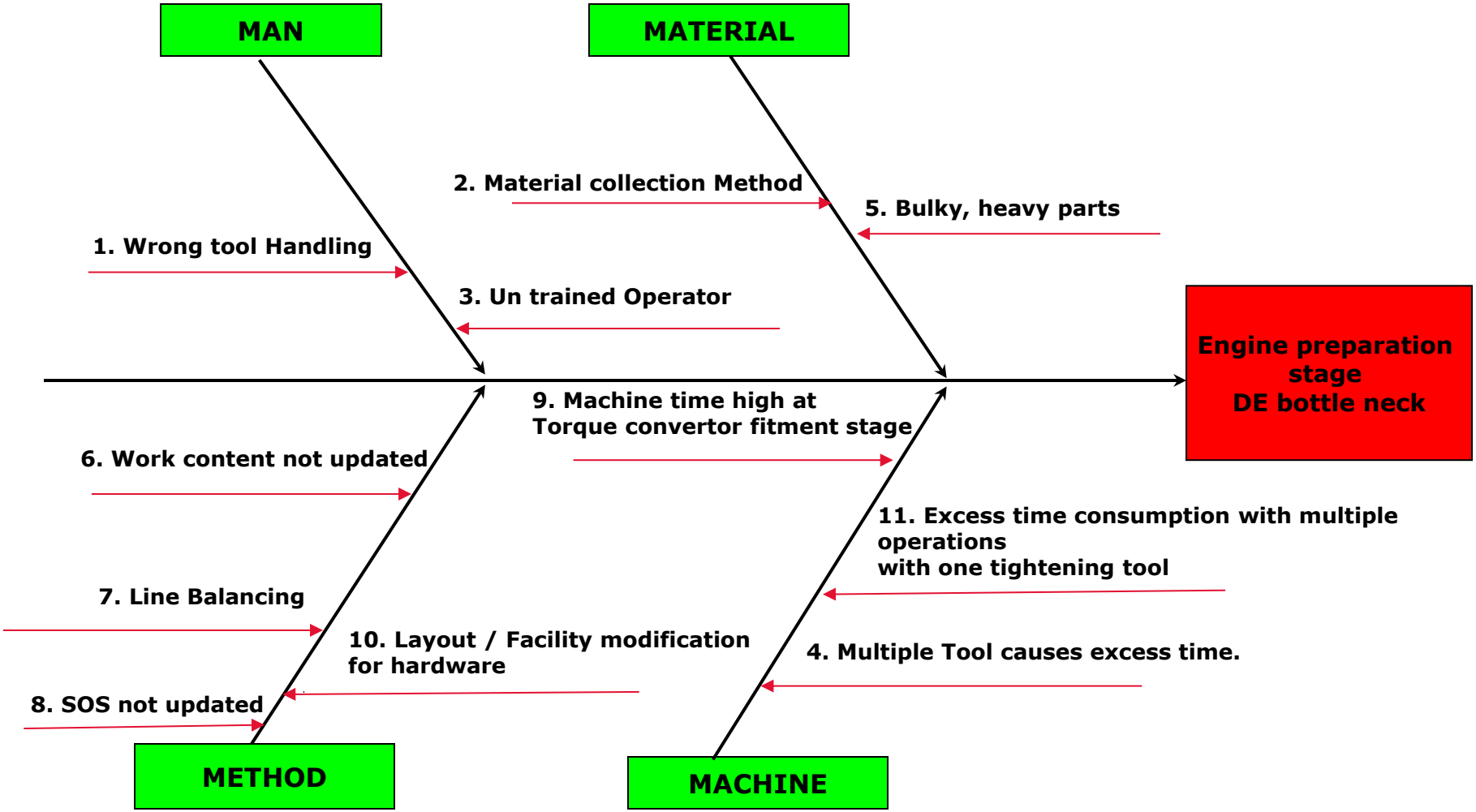
# Step 5. Analysis

**Brainstorming - List of Possible Causes**

| S.NO | Possible causes |
|---|---|
| 1 | Wrong tool handling |
| 2 | Material collection Method |
| 3 | Untrained Associate |
| 4 | Multiple Tool / operation causes excess time. |
| 5 | Bulky, Heavy parts |
| 6 | Work content not updated |
| 7 | Line balancing |
| 8 | SOS not updated |
| 9 | Machine time high at Torque convertor fitment stage |
| 10 | Layout / Facility modification for hardware |
| 11 | Excess time consumption with multiple operations with one tightening tool |

# Step 5. Analysis

## 5.1 Possible Causes :-



# Step 5. Analysis

## 5.1 Identify possible causes: Brainstorming

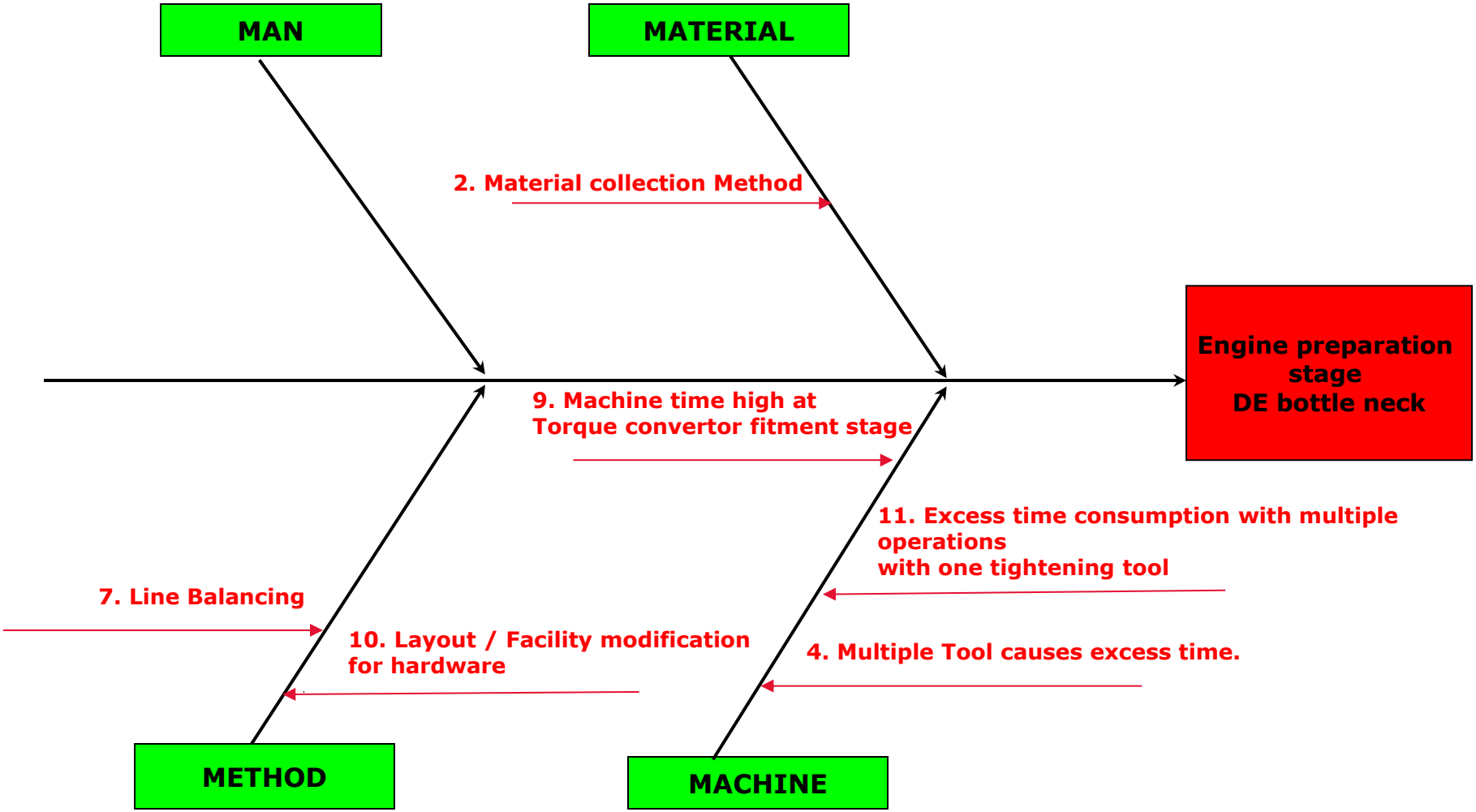
| Sr. no | Possible Causes | Nikhil Hatedkar | Rohit Ghuge | Tushar Patil | Rajesh Dhanokar | Jitendra Bhambre | Pravin Patil | Total |
|---|---|---|---|---|---|---|---|---|
| 1 | Wrong Tool handling | 3 | 1 | 0 | 0 | 1 | 3 | 8 |
| 2 | Material collection method | 9 | 9 | 3 | 9 | 9 | 3 | 42 |
| 3 | Untrained Associates | 0 | 0 | 1 | 1 | 3 | 9 | 14 |
| 4 | Multiple Tool / operation causes excess time. | 9 | 9 | 9 | 6 | 9 | 3 | 48 |
| 5 | Bulky , Heavy parts | 1 | 0 | 1 | 1 | 0 | 9 | 12 |
| 6 | Work content not updated | 0 | 0 | 1 | 1 | 1 | 3 | 6 |
| 7 | Line Balancing | 9 | 9 | 1 | 9 | 9 | 3 | 40 |
| 8 | SOS not updated | 0 | 1 | 3 | 3 | 1 | 9 | 17 |
| 9 | Machine time high at Torque convertor fitment stage | 9 | 9 | 9 | 1 | 9 | 9 | 46 |
| 10 | Layout / Facility modification for hardware | 9 | 3 | 3 | 9 | 9 | 9 | 42 |
| 11 | Excess time consumption with multiple operations with one tightening tool | 3 | 9 | 9 | 3 | 9 | 9 | 42 |

**Scale: 0=None, 1=Low, 3=Moderate, 9=Strong**

**Threshold Limit = 40**

# Step 5. Analysis

## 5.1 Probable Causes :-



# Step 5. Analysis

**List of causes not considered as probable causes**

| SNO | Not considered Causes | Reasons |
|---|---|---|
| 1 | Wrong Tool handling | Trained & Regular Operator |
| 3 | Untrained Associates | Trained & Regular Operator |
| 5 | Bulky , Heavy parts | As per Ergonomics study |
| 6 | Work content not updated | Work Content Signed off |
| 8 | SOS not updated | SOS is dynamic documents |

# 5.1 Identify Probable cause

| SR NO | List of Probable Causes |
|---|---|
| **2** | Material collection method |
| **4** | Multiple operations causes excess time. |
| **7** | Line Balancing |
| **9** | Machine time high at<br>Torque convertor fitment stage |
| **10** | Layout / Facility modification for hardware |
| **11** | Excess time consumption with multiple operations with one tightening tool |

# Step 5.2 Test of Hypothesis

**Confirm the probable cause:**

| No | Probable Cause | Testing method | Observations | Conclusion |
|---|---|---|---|---|
| 2 | Material collection method | By visual | On Gemba it is observed that Associates are using shortcuts to complete the task | **Hypothesis Valid** |

## Steps (NVA) for pre-assembly of W112 AT vehicle -

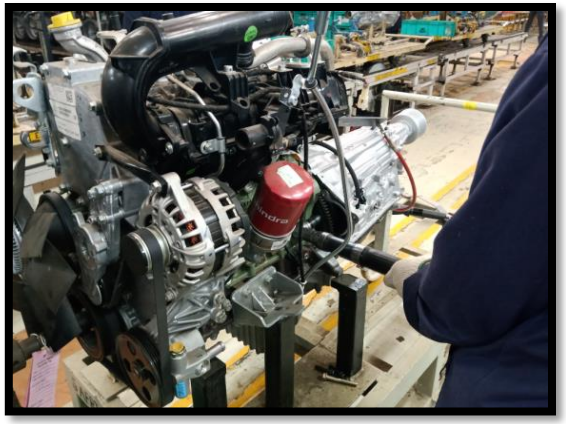
1) Steps (A) – Engine move to offline assy. Area
2) Steps (B) – Gear box move to offline assy. Area
3) Steps (C) – alignment of engine and indexing fixture
4) Steps (D) – Tightening of torque converter using electrical tool
5) Steps (E) – Engine move towards conveyor and place assemble engine on conveyor

Inference- Out Of 20 Vehicle, Associate follows SOS in each vehicle , but associate require no. of stage during sub assembly of W112 AT engine so it is valid cause

# Step 5.2 Test of Hypothesis

**Confirm the probable cause:**

| No | Probable Cause | Testing method | Observations | Conclusion |
|---|---|---|---|---|
| 4 | Multiple operations for fitment | By visual | For each cycle for tightening of converter plate mounting bolt associate manually press inching button | **Hypothesis Valid** |



Indexing machine installation

Press inching button 7 times

Converter plate fitment

**Runner RPM is also low**

**Inference-** Associate press more no. of time inching button for tightening converter plate mounting bolts

# Step 5.2 Test of Hypothesis

**Confirm the probable cause:**

| No | Probable Cause | Testing method | Observations | Conclusion |
|---|---|---|---|---|
| 7 | Line Balancing . | By work contained sheet | Group Associate wise balancing and found that some Associate are more utilized and some are less utilized | **Hypothesis Valid** |

| Vehicle Name | Activity |
|---|---|
| W112_AT | GEAR BOX LIFTING FROM STORAGE AREA |
| W112_AT | GEAR BOX DROP ON ASSEMBLY TROLLEY |
| W112_AT | GEAR BOX DECK ON ENGINE HOUSING |
| W112_AT | ENGINE ALIGNMENT WITH INDEXING FIXTURE |
| W112_AT | GEAR BOX LOCKING PATTI REMOVAL _AT |
| W112_AT | FLEX PLATE TO TORQUE CONVERTER 6 BOLT FITMENT AND TIGHTENIN |
| W112_AT | WATER COOLING LINE CP CLAMP CONNECTION 2 NOS WITH ENGINE _A |
| W112_AT | BREATHER PIPE ROUTINE AND CLIPPING ON METALLIC PIPE_AT |
| W112_AT | OIL COOLING LINE FITMENT ON TRANSSION 2 BENJO BOLT LOOSE FITMENT_AT |
| W112_AT | AT TRANSMISSION WIRING CLIPPING BKT_AT |
| W112_AT | AT TRANSMISSION WIRING FITMENT LH SIDE_AT |
| W112_AT | STARTER MOTOR DUST SHIELD ADDITIONAL PART FITMENT_AT |
| W112_AT | ENGINE FEED ON CONVEYOR |

Offline associate are more operation

Operator Utilization more than 100%

Inference- As per work contain sheet we found some group associate wise are more utilized , so it is valid cause

# Step 5.2 Test of Hypothesis

## Confirm the probable cause:

| No | Probable Cause | Testing method | Observations | Conclusion |
|---|---|---|---|---|
| 9 | Machine time high at convertor plate fitment stage | By visual | Indexing fixture time is too high | **Hypothesis Valid** |

| SR NO | Chassis line critical process | Rank | Stages | Machine time in Sec. | | | | | Max in sec | Model / Remarks |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Rear Axle Compression fixture | A | Axle | 79 | 77 | 78 | 79 | 79 | 79 | Marazzo |
| 2 | Front Axle Compression Fixture | A | IFS | 61 | 65 | 62 | 61 | 61 | 65 | Scorpio |
| 3 | Brake leak test | A | Engine drop | 54 | 54 | 54 | 54 | 54 | 54 | Marazzo |
| 4 | Chassis no. punching | A | Engine drop | 68 | 64 | 65 | 63 | 66 | 68 | Marazzo |
| 5 | Convertor plate tightening machine | A | Engine Preparation | 98 | 96 | 97 | 98 | 97 | 98 | W112 - AT |
| 6 | Tyre multispindle | B | Tyre & oil | 52 | 51 | 53 | 51 | 54 | 54 | Marazzo |
| 6 | Transmission oil filling | B | Tyre & oil | 62 | 64 | 62 | 63 | 64 | 64 | Marazzo |

Machine time Constraint for 183 Setups

Inference- As per work contain sheet we found machine time is more compare to actual activity time , hence it is valid cause

# Step 5.2 Test of Hypothesis

## Confirm the probable cause:

| No | Probable Cause | Testing method | Observations | Conclusion |
|---|---|---|---|---|
| 11 | Layout / Facility modification for hardware | By visual | loose fitment of converter plate hardware are kept in next stage | **Hypothesis Valid** |

**More no. of steps for Pre assembly of W112 AT**

Inference- Out Of 20 Vehicle, Associate more steps require for pick hardware, hence it is a valid cause.

# Step 5.2 Test of Hypothesis

## Confirm the probable cause:

| No | Probable Cause | Testing method | Observations | Conclusion |
|---|---|---|---|---|
| 12 | Excess time consumption with multiple operations with one tightening tool | By visual and Gamba | As per visual inspection there is no multiple tightening operation with single runner | **In Valid** |

W112 - AT - Subassy Operations.

| SR NO. | Operations | Runner available |
|---|---|---|
| 01) | Bellhousing tight - | available |
| 02) | Flywheel tightening - | available |
| 03) | torque convertor fit - | available |
| 04) | propeller shaft cap - | available |
| | gear box insulator - | available |

**Separate tool provided For all operations**

Inference- Out Of 10 Vehicle, Associate tightening & fitment parts with separate runner

# TEST OF HYPOTHESIS

## 5.2 Confirm the probable cause:

| No | Probable Cause | Testing method | Conclusion |
|---|---|---|---|
| 2 | Material collection method | By visual | **Hypothesis valid** |
| 4 | Multiple operations causes excess time. | By machine time with activity time | **Hypothesis valid** |
| 7 | Line Balancing | By work contained sheet | **Hypothesis valid** |
| 9 | Machine time high at Torque convertor fitment stage | By work contained sheet | **Hypothesis valid** |
| 10 | Layout / Facility modification for hardware | By visual | **Hypothesis valid** |
| 11 | Excess time consumption with multiple operations with one tightening tool | By visual | **In Valid** |

# WHY - WHY Analysis

## 5.3 Identify the root causes:

| No | Valid Probable Cause | WHY | WHY | WHY | ACTION / COUNTERMEASURE |
|---|---|---|---|---|---|
| 4 | Material collection method | Gear box and Engine move to offline area for assy. | Indexing machine is only at offline | No space for indexing machine at online | Indexing machine shift to online |
| 7 | Multiple operations causes excess time. | For tightening of converter plate to engine | Require sub assy. for indexing machine | For tightening of converter plate require before engine assy. | Gear box decking with engine operation shift to online |
| 9 | Line Balancing | Dependent operation | Require offline operation | Indexing machine at offline | Engine wiring operation shifted to Clutch and pressure fitment stage & starter motor fitment operation fitment to engine wiring |
| 10 | Machine time high at Torque convertor fitment stage | Low RPM tool use for tightening of mounting bolt | Inching require for mounting bolt tightening | Require rotate to converter plate bolt fitment | Manual inching require for each bolt tightening |
| 11 | Layout / Facility modification for hardware | Hardware rack is too long | There is moving other parts moving trolley acquire large space | | Hardware and material bin next stage |

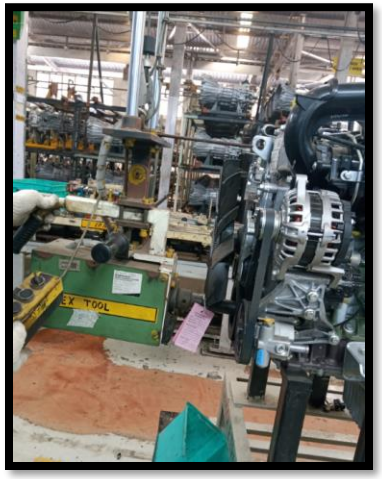
# 6. Implement the corrective measures

## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 4 | Indexing machine shift to online | Provide center rail with balancer and install machine | 10.08.19 | Rohit Ghuge | CLOSED |

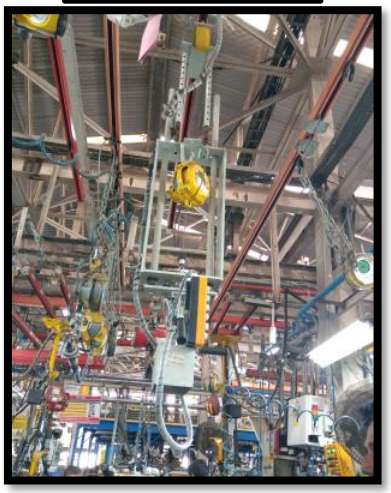
**Shifting of indexing machine at online**

**BEFORE**



Torque converter tightening indexing machine at offline stage

**AFTER**



Indexing machine shifted to online area by, providing central rail

# Step 6. Implement Action

## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 7 | Multiple operations causes excess time. | Provide separate lifting tackle for install gear box to engine | 24.7.19 | Rohit Ghuge / Nikhil | CLOSED |

BEFORE

AFTER

1. Activity shifted to Bellhousing decking stage
2. By provide special lifting tackle for decking

# Step 6. Implement Action

## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 9 | Line Balancing | Engine wiring operation shifted to Clutch and pressure fitment stage & starter motor fitment operation fitment to engine wiring | 13.11.19 | Nikhil Hatedkar / Rajesh Dhanokar | CLOSED |

| SN | Stage Name | Work |
|---|---|---|
| 1 | Clutch and Pressure plate fitment | No work / at line |
| 2 | Gear box decking stage | No work / at line |
| 3 | Stator motor fitment | Startor fitment / at line |
| 4 | Engine wiring stage | Engine wiring / at line |
| 5 | Torque converter fitment | Offline |

| SN | Stage Name | Work |
|---|---|---|
| 1 | Clutch and Pressure plate fitment | Engine wiring fitment |
| 2 | Gear box decking stage | Gear box decking with engine |
| 3 | Startor motor fitment | Torque converter fitment |
| 4 | Engine wiring stage | Startor motor |
| 5 | Torque converter fit | Delet from offline |

**Before- 1) Clutch and pressure plate associate no work for W112 AT**
**2) Gear box decking associate no work for W112 AT**

**After – 1) At Clutch and Pressure plate fitment stage – Wiring activity added**
**2) Decking stage – Gear box decking activity added**
**3) At wiring stage – Startor motor activity shifted**

# Step 6. Implement Action

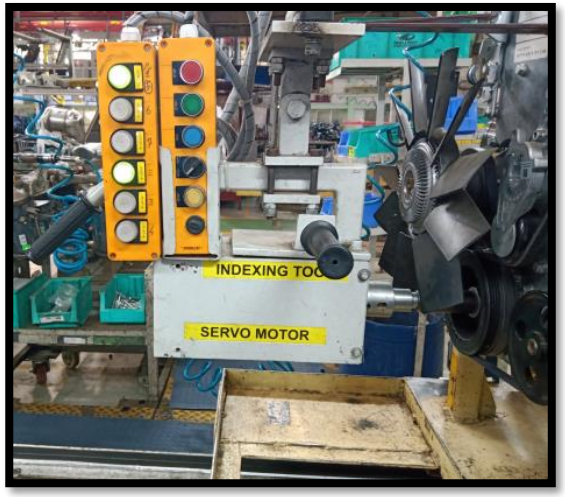
## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 10 | Manual inching require for each bolt tightening | Inching time reduce by using Auto inching with Poka-yoke | 16.11.19 | Rohit Ghuge / Tushar Patil | CLOSED |

**BEFORE**

1. After every bolt tightening of torque converter mounting bolt there is require inching for every cycle

**AFTER**




1.Automatically converter rotate after bolt tightening

Elimination of NVA's

# Step 6. Implement Action

## 6.1 Actions Decided

| No | Root Cause | Action plan | Actual date | Responsibility | Status |
|---|---|---|---|---|---|
| 10 | Hardware and material bin next stage | Hardware and material bin provided near at stage | 14.09.19 | Nikhil Hatedkar | CLOSED |

BEFORE

Mounting bolt and runner parking is too long from actual fitment stage

AFTER

Modification done at stage and hardware and material bin location changed

Elimination of NVA's

# 6. Implement the corrective measures

## Possible Remedies Identified

Six possible solutions identified for Engine drop bottle neck issue

| Legends | |
|---|---|
| High | Low |

| S N | Root Cause | Remedy Developed | Time to implement | Cost of Improvement | Easily reversible | Complexity | Decision |
|---|---|---|---|---|---|---|---|
| 4 | Indexing machine shift at offline | Provide center rail with balancer and install machine | Low | Medium | Medium | Medium | Accepted |
| 7 | Gear box decking with engine operation shift to online | Provide separate lifting tackle for install gear box to engine . | Low | Low | Low | Medium | Accepted |
| 9 | Line Balancing | Engine wiring operation shifted to Clutch and pressure fitment stage & starter motor fitment operation fitment to engine wiring | Low | Medium | High | Low | Accepted |
| 10 | Manual inching require for each bolt tightening | Inching time reduce by using Auto inching with Pokayoke | Medium | Low | Low | Low | Accepted |
| 11 | Hardware and material collection provide close to associate | Existing trolley modification and space to be created for W112 AT hardware parts | Low | Medium | High | Low | Accepted |

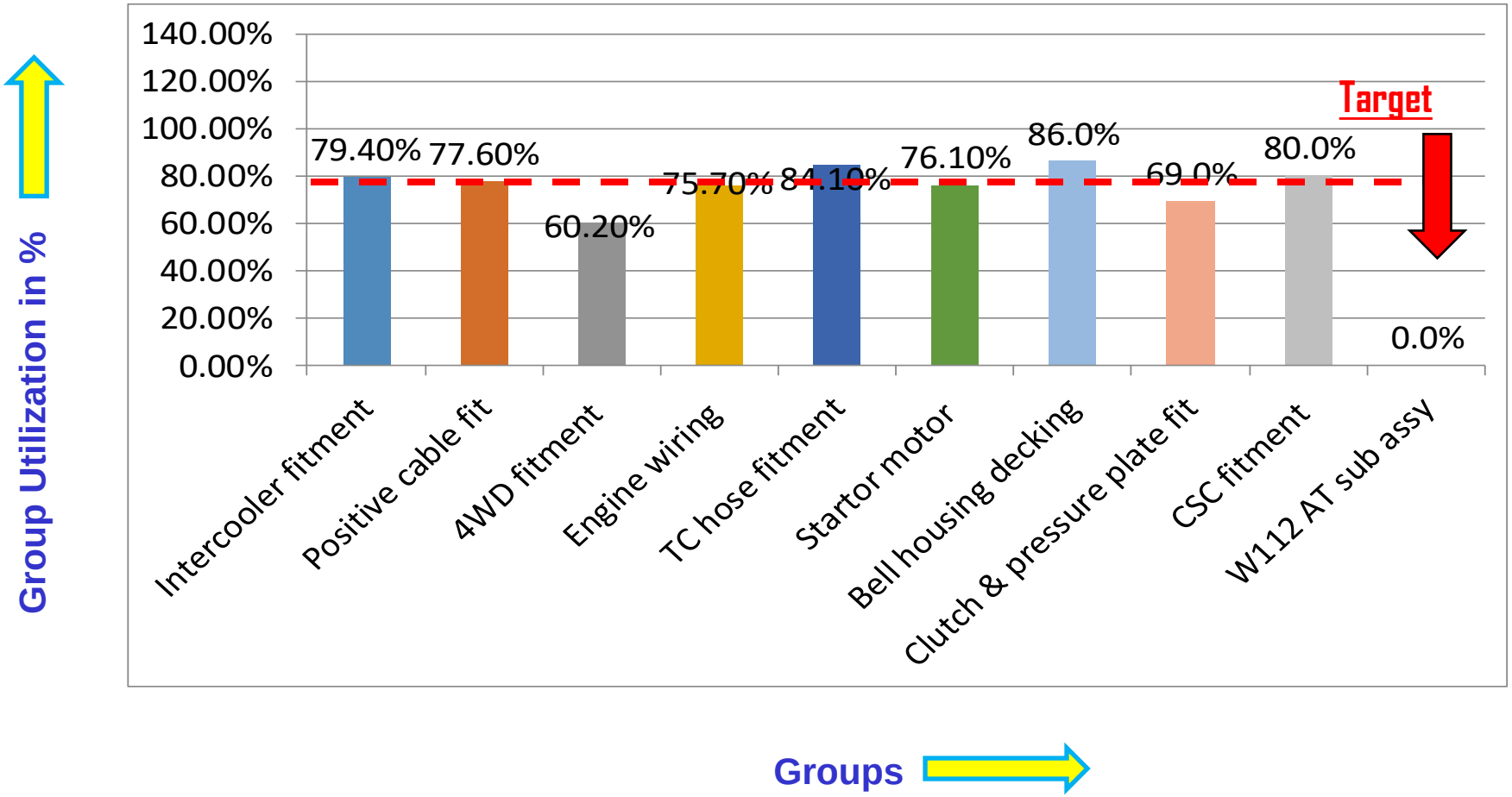
# Step 6. Implement Action

## Action Plan & Status

| SNo | Proposed Ideas for Reduction in Work content | Target | Responsibility | Status |
|---|---|---|---|---|
| 1 | Provide center rail with balancer and install machine | 10.08.19 | Rohit Ghuge / Tushar Patil | Completed |
| 2 | **Process Flow Change**<br>**a.** Wiring operation shift to clutch and pressure plate stage<br>**b.** Offline decking activity shift at online<br>**c.** Starter motor fitment operation shift to wiring fitment stage | 13.11.19 | N. HATEDKAR | Completed |
| 3 | Gear box decking with engine operation shift to online | 24.07.19 | Tushar Patil / N. HATEDKAR | Completed |
| 4 | Hardware and material collection provide close to associate | 14.09.19 | N. HATEDKAR | Completed |
| 5 | Work Content Signoff | NA | N. HATEDKAR | Completed |

# Step 7. Confirm Effect /Verify Results

**Chassis line Utilization __ with W112 AT vehicle**



# 7. Check & confirm the results & effects

## 7.2 Full Scale Implementation plan:

| SN | Action Taken | Actual date | Responsibility | Status |
|---|---|---|---|---|
| 1 | Provide center rail with balancer and install machine | 10.08.2019 | Rohit Ghuge / Tushar Patil | Close |
| 2 | Gear box decking with engine operation shift to online | 24.07.2019 | Tushar Patil / N. HATEDKAR | Close |
| 3 | Process Flow Change | 13.11.2019 | N. HATEDKAR | Close |
| 4 | Machine inching time reduce by Poka-yoke | 16.11.2019 | N. HATEDKAR | Close |
| 5 | Hardware and material collection provide close to associate | 14.09.2019 | N. HATEDKAR | Close |

# Step 7. Confirm Effect /Verify Results

## 7.3 Tangible & Intangible benefits:

### Tangible Benefits

1. By NVA Reduction Process flow change save 2 man
2. Implementation Without any increase in manpower
3. Space saving by 150 sq. feet

### Intangible Benefits

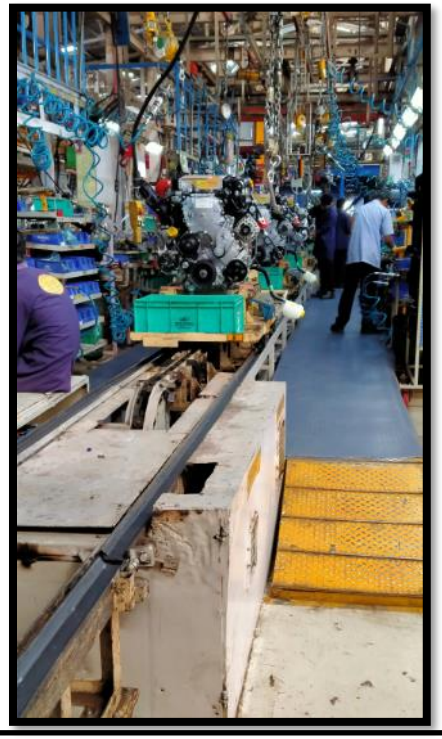
1. Meeting Market requirement of W112 AT Vehicle

# Step 6. Implement Action

BEFORE

Before- we assemble W112 AT vehicle converter plate and gear box decking at offline area

AFTER



After – W112 AT pre assembly shifted to online

Space saving by 150 Sq. feet

# 8. Standardization

## 8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

| SN | Document | Actual date | Responsibility | Status |
|---|---|---|---|---|
| 1 | Work contain sheet | 14.09.2019 | Nikhil HATEDKAR | Close |
| 2 | Process flow | 13.11.2019 | Nikhil HATEDKAR | Close |
| 3 | Most element sheet | 16.11.2019 | Nikhil HATEDKAR | Close |

## 8.2 Training:

| SN | Training Identified | Actual date | Responsibility | Status |
|---|---|---|---|---|
| 1 | Model training done | 20.12.2019 | Nikhil HATEDKAR | Close |

## 8.3 Additional Audit & check

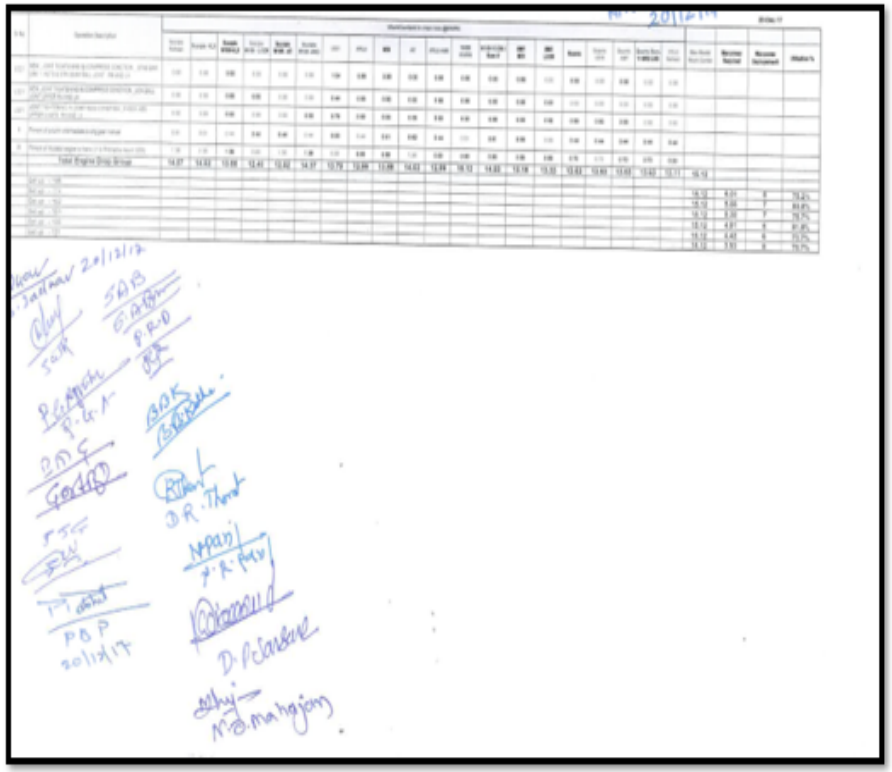
| | | | | |
|---|---|---|---|---|
| | NIL | | | |

# Step 8. Standardization

## 8.1 Updation of documents:

### 1. Work content Sheet

**From:** DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
**Sent:** 14 Sep 2019 15:35
**To:** HATEDKAR NIKHIL D <HATEDKAR.NIKHIL@mahindra.com>
**Subject:** Engine Preparation WC
Pl find attached file of Wc



# Step 8. Standardization

## 8.1 Updation of documents:

## 2. Process Flow Chart

| YLO TCF | Next Aggregate No.: | Approved by : Sandeep Joshi | Rev. Date : 16.06.15 | PFC Date(Orig) : 10 Nov. 2005 |
|---|---|---|---|---|
| Module Name : CHASSIS | Legends/ Abbrevations | Critical Char: <CC> | Transportation | Storage | Operation / Assembly | Inspection |
| Stage No.: | | Significant Char : <SC> | Operation & Inspection | Delay | Incoming Parts | |

| Opn No | Work Stn No / Machine No | Opn. Description | Main Assly Flow | Offline Assly Flow | Product Characteristics | Process Characteristics | Char | Open issues/ Remarks |
|---|---|---|---|---|---|---|---|---|
| 5790 | NSC12 | Fitment Of Clutch Fork & Release Bearing On Transmission (DIESEL Only ) | | | > Fitment of bearing on the clutch fork | > Locking of fork over transmission ball stud<br>> Even application of greese | | |
| ~~5800~~ | ~~NSC12~~ | ~~Fitment Of Clutch Fork (CSC) On Transmission (PETROL)~~ | | | ~~> Torque:9 ± 2 Nm~~ | ~~> Calibration of nut runner~~ | ~~CC~~ | operation deleted |
| 5801 | NSC12 | Fitment of clutch Fork(CSC) on transmission(Eagle) | | | > Torque - 9 +/- 2 Nm (Army 3 point)<br>> Torque -25 +/-3 nm (W105- 2 point) | > Fitment of bleeder on clutch fork<br>> Simultaneous Tightening<br>> Tightening in two steps 1. 2Nm and 2. 9Nm<br>> Printout Confirmation of Torque delivered<br>>Use of teflon sleeve for ensuring correct alignment | CC | |
| 5802 | NSC12 | Fitment of Dust Cover on CSC | | | >Locking of Dust Cover on CSC. | > Positive Locking | | |
| | | | | | | > Complete locking of clutch tube | | |

**Activity shifted to another stage**

# Step 8. Standardization

## 8.1 Updation of documents:

## 2. PFMEA

| | | | |
|---|---|---|---|
| Plant: **Nasik** | Model: **Scorpio** | *Mahindra & Mahindra Ltd.* | *Cross Functional Team* |
| Part No/s.: | Part Name/Process: | | Nikhil / Suryabhan; Hari Nagar |
| Latest Change Letter: **NA** | Optn No.: **6800** | *Process Potenial Failure Mode & Effect Analysis.* | Vikrant Ghule; Vikrant Marade |
| PU/Shop: **Scorpio** | Optn Descptn: **Fitment of convertor plate on engine at online** | Reviewed by: **Nikhil Hatedkar** / Rev. No: **25** | Doc No.: **PFMEA/SPU/SCORPIO/6800** |
| Module/Cell: **TCF- Chassis** | **M/C Name & No.:** | Approved by: **Taj Sheikh** / Rev. Date: **5.8.19** | FMEA Date (Orig): **12/01/2002** |

| Process Function | Requirements | Potential Failure Mode | Potential Effect(s) of Failures | Sev | Class | Potential Cause(s)/Mechanism(s) of Failures | Current Process Controls Prevention | Occur | Current Process Controls Detection | Detect | RPN | Recommended Action(s) | Responsibility & Target Completion Date | Action Results: Action Taken | Sev | Occ | Det | RPN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Torque convertor plate fitment | tightneing of convertor plate on engine | 1) bolt under torqued/loose. | **Mfg & Assy :** 1) Rattle noise and rework. (5) **End User :** 1) Oil leakage will hamper clutch operation.(8) | 9 | cc | 1) Operator does not tighten the Nut. | Pokayoke provided with torque sticker | 2 | Torque value print checking at chassis buyoff. | 7 | 126 | If bolt remain loose engine conveyor get stop. | PP/TP | Action completed. | | | | |
| | | | | | | 2) Improper power tool calibration | Calibration as per frequency decided in control plan. Torque Setting is Checked on controller | 2 | Sticker for due date for calibration on tool. | 9 | 162 | | | | | | | |
| | | 2) Tightening sequence not follow | **Mfg & Assy :** 1) Rework. **End User :** 1) Oil leakage will hamper clutch operation.(8) | 9 | cc | 1) I Operator does not follow proper sequence during tightening | Auto inching facility provided for auto sequence | 2 | Auto inching facility provided for auto sequence | 5 | 90 | | | | | | | |

| Rev No.: | Rev Date | Revision Summary | Reason for Revision |
|---|---|---|---|
| 1 | 27.05.06 | No change | Review for 2DS. |
| 2 | 28/10/06 | No change. | Document reviewed. |
| 3 | 12-07-2007 | No change. | Document reviewed. |
| 4 | 12-06-2008 | No change. | Document reviewed. |
| 5 | 09-10-2009 | No change. | Document reviewed as per 4th edition. |
| 6 | 05.08.09 | No change. | Document reviewed. |
| 7 | 04.02.10 | No change. | Document reviewed. |
| 8 | 30.10.11 | No change. | Document reviewed. |
| 9 | 24.03.12 | No change. | Document reviewed. |
| 10 | 22.07.12 | Detection revised for CSC Print out started from 9 to 7. | Document revised for CSC pokayoke & Print out started for CSC. |
| 11 | 20.09.12 | No change. | Document reviewed. |
| 12 | 29.04.13 | No change. | Six monthly review |
| 13 | 31.12.13 | No change. | Six monthly review |
| 14 | 20.06.14 | Conveyor installation at engine preparation. | Pokayoke link with conveyor. |
| 15 | 17.10.14 | No change. | Six monthly review |
| 16 | 29.05.15 | No change. | Six monthly review |
| 17 | 30.11.15 | No change. | Six monthly review |
| 18 | 30.04.16 | No change. | Six monthly review |
| 19 | 26.10.16 | No change. | Six monthly review |
| 20 | 25.04.17 | No change. | Six monthly review |
| 21 | 13.10.17 | No change. | Six monthly review |

# Step 8. Standardization

## 8.2 Training:

| S.NO | Training Description | No. of operators | Trainer | Date |
|---|---|---|---|---|
| 1 | Model Training | 48 | Sanjay Patil 97541 | 20.12.2019 |
| 2 | Most element sheet | 20 | Nikhil H 23061163 | 16.11.2019 |
| 3 | Work Content Sheet | 20 | Nikhil H 23061163 | 14.09.2019 |

# Step 8. Standardization

## 8.3 Additional Audit / Check

### Work content sign off made compulsory for Project / EN

| PROJECT/EN DELIVARABLES | |
|---|---|
| CHANGE DESCRIPTION | |
| REASON | |
| OWNER | |
| CDMM | |
| EXP. IMPL. DATE | |

| R N | BUILD DELIVERABLES | DETAILS | C/O | lical | GRY | Remarks |
|---|---|---|---|---|---|---|
| 1 | Copmpatibility with 220 UPS (127 secs TACT) | | | | | |
| 1 | STATIC BUILD | | | | | |
| 2 | BOM VERIFICATION | | | | | |
| 3 | WORK CONTENT ESTIMATION | | | | | |
| 4 | IDENTIFICATION OF FTG | | | | | |
| 5 | OPERATOR TRAINING | | | | | |
| 6 | CONCERNS IDENTIFICATION | | | | | |
| 7 | COMMUNICATION OF CONCERNS TO THE DESIGNER | | | | | |
| 8 | PROCESS FLOW | | | | | |
| 9 | PROCESS SHEET | | | | | |
| 10 | PROCESS FMEA | | | | | |
| 11 | CONTROL PLAN | | | | | |
| 12 | SOS VALIDATION | | | | | |
| 13 | SOS UPDATION | | | | | |
| 14 | TPPS STUDY | | | | | |
| 15 | ANOVA-C AND PRODUCT VALIDATION | | | | | |
| 16 | ES TEST | | | | | |
| 17 | REPAIR STANDARD | | | | | |
| 18 | TRAVELLER CARD | | | | | |
| 19 | BUILD SHEET | | | | | |
| 20 | BUY OFF CHECK SHEET | | | | | |
| 21 | MSDS | | | | | |
| 22 | HIRA | | | | | |

# 9. Summarize & Plan future actions

## 9.1 Conclusion

Smooth Implementation of Marazzo

1. Effective Brainstorming with CFT
2. Discussion with Line Associates for their Suggestion

# 9. Summarize & Plan future actions

**9.2 Horizontal Deployment:**

For Upcoming Models:

1. Scorpio (W150) – BS6 - 6
2. Marazzo (U325)- BS6 Vehicle
3. Thar (W501) – Diesel and Petrol automatic transmission

# 9. Summarize & Plan future actions

## 9.3 Lessons Learnt:

### TGW (Things Gone Wrong) :

Tooling / FTG need identification got delayed due to model variation.

### TGR (Things Gone Right) :

1. Converting the manual operations into automatic & eliminating the NVA's we are able to implement W112 AT 100% online.

# Thank You!